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# Free variables and basis for $N(A)$

Let $A \in \mathbb{F}^{m\times n}$ be a matrix in reduced row-echelon form. Recall that we can get all the solutions to $Ax = 0$ by setting the free variables to distinct parameters. Then the set of solutions can be written as a linear combination of $n$-tuples where the parameters are the scalars. These $n$-tuples give a basis for the nullspace of $A$. Hence, the dimension of the nullspace of $A$, called the **nullity of** $A$, is given by the number of non-pivot columns.

We now look at an example of finding a basis for $N(A)$. Let $A \in \mathbb{R}^{2\times 4}$ be given by $\begin{bmatrix} 1 & -1 & -1 & 3 \\ 2 & -2 & 0 & 4 \end{bmatrix}$. We perform the following elementary row operations:

$$\begin{aligned}
& \begin{bmatrix} 1 & -1 & -1 & 3 \\ 2 & -2 & 0 & 4 \end{bmatrix} \\
\xrightarrow{R_2 \leftarrow R_2 - 2R_1} & \begin{bmatrix} 1 & -1 & -1 & 3 \\ 0 & 0 & 2 & -2 \end{bmatrix} \\
\xrightarrow{R_2 \leftarrow \frac{1}{2}R_2} & \begin{bmatrix} 1 & -1 & -1 & 3 \\ 0 & 0 & 2 & -2 \end{bmatrix} \\
\xrightarrow{R_2 \leftarrow \frac{1}{2}R_2} & \begin{bmatrix} 1 & -1 & -1 & 3 \\ 0 & 0 & 1 & -1 \end{bmatrix} \\
\xrightarrow{R_1 \leftarrow R_1 + R_2} & \begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{bmatrix}
\end{aligned}$$

To obtain all solutions to $Ax = 0$, note that $x_2$ and $x_4$ are the free variables. Set $x_2 = s$ and $x_4 = t$. Then all the solutions are given by

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} s - 2t \\ s \\ t \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

Hence, a basis for $N(A)$ is given by $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$ and $\dim(N(A)) = 2$.

**Remark:** Note that one could have obtained directly the first vector by setting $x_2 = 1$, $x_4 = 0$ and then solving for $x_1$ and $x_3$ and the second vector by setting $x_2 = 0$, $x_4 = 1$ and then solving for $x_1$ and $x_3$. In general, if $A$ is in RREF, then a basis for the nullspace of $A$ can be built up by doing the following: For each free variable, set it to 1 and the rest of the free variables to zero and solve for the pivot variables. The resulting solution will give a vector to be included in the basis.

Quick Quiz

# Exercises

For each of the matrices, find a basis for its nullspace.

1. $A \in \mathbb{R}^{3\times 4}$ given by $\begin{bmatrix} 1 & -1 & 0 & 1 \\ 0 & 2 & 1 & -1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$. Show answer

2. $A \in GF(2)^{2\times 4}$ given by $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$. Recall that $GF(2)$ is the field consisting of only 0 and 1 with the usual integer addition and multiplication except that "$1+1=0$". Show answer